

Irrationality after One Stabilization of Universally CH_0 -Trivial Cubic Threefolds

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Abstract

Does universal CH_0 -triviality force rationality after stabilization? We construct a non-isotrivial one-parameter family of smooth cubic threefolds X_b for which X_b is universally CH_0 -trivial but $X_b \times \mathbf{P}^1$ is irrational. More generally, we prove that $X \times \mathbf{P}^1$ is irrational for every smooth cubic threefold X .

The cycle argument proves a local cofactor-saturation theorem for principally polarized elliptic-power quotients; semisimple graph slopes are a sufficient mechanism. Six elliptic quotients of the intermediate Jacobians in the nonstandard A_5 -invariant cubic pencil satisfy its hypotheses. Voisin’s theorem then gives universal CH_0 -triviality fibre by fibre.

For irrationality, we count primitive-sixth eigenvalues of framed formal monodromy on the numerical small even quantum connection. Iritani’s blowup formula transports the count through weak factorization; a divisor-tagging argument handles its possibly noninjective center specialization. The count is nonzero for a cubic and zero for every point, curve, and surface. Thus the cycle-theoretic and quantum detectors diverge on the same smooth fourfolds.

Keywords. cubic threefold; universal CH_0 -triviality; irrationality; intermediate Jacobian; quantum connection; weak factorization.

1 Introduction

A smooth projective variety is universally CH_0 -trivial if the degree map

$$\deg : CH_0(Y_L) \longrightarrow \mathbf{Z}$$

is an isomorphism for every field extension $L/\mathbf{C}$. This condition is a central consequence of stable rationality, and its failure is a standard stable-rationality obstruction. Its validity, however, need not provide a rational parametrization. We ask how far that gap can persist after one explicit stabilization.

Our examples are fourfolds of the form $Y = X \times \mathbf{P}^1$, where X is a smooth cubic threefold in the nonstandard A_5 -invariant pencil. Two independent mechanisms act on the same Y . The intermediate Jacobian of X has an algebraic primitive minimal class; this gives universal CH_0 -triviality. The small even quantum connection of every smooth cubic has primitive-sixth framed formal monodromy; Iritani’s quantum blowup formula shows that its multiplicity survives one stabilization.

Here “nonstandard” has a precise representation-theoretic meaning. Let W_5 be the irreducible five-dimensional complex representation of A_5 , with character $(5, 1, -1, 0, 0)$ on the classes $(1, 2A, 3A, 5A, 5B)$. The symmetric-cube character formula gives

$$\dim(\mathrm{Sym}^3 W_5^*)^{A_5} = 2.$$

Thus $\mathbf{P}((\mathrm{Sym}^3 W_5^*)^{A_5}) \cong \mathbf{P}^1$ is a pencil of cubic hypersurfaces in $\mathbf{P}(W_5)$. We denote its smooth locus by B° . It is the complement of the discriminant in the pencil and is connected. This is Hartlieb’s irreducible-character component M_{H_1} , as opposed to the component coming from the reducible permutation representation $\mathbf{1} \oplus W_4$ [7, Lemma 5.5].

Theorem 1.1 (Separation after one stabilization). *Let $\mathcal{X} \rightarrow B^\circ$ be the smooth family in the nonstandard A_5 -invariant pencil of cubic threefolds. For every $b \in B^\circ(\mathbf{C})$,*

$\mathcal{X}_b$ is universally CH_0 -trivial, $\mathcal{X}_b \times \mathbf{P}^1$ is universally CH_0 -trivial and irrational.

The family is non-isotrivial in coarse moduli.

The irrationality statement does not depend on the symmetry of the pencil.

Theorem 1.2 (One-step irrationality). *For every smooth complex cubic threefold X , the fourfold $X \times \mathbf{P}^1$ is irrational.*

We do not claim that X is stably irrational. From the second stabilization onward, weak-factorization centers can themselves have cubic dimension, and the low-dimensional exclusion used here no longer applies.

How the two proofs meet

The cycle-theoretic half starts from six elliptic quotients of the Fano surface. Their Rosati Gram matrix is $6I - J$, so any five axes give an isogeny $E^5 \rightarrow J(X)$ of polarization type $(1, 6, 6, 6, 6)$. The two-primary principal kernels form $\mathbf{P}^1(\mathbf{F}_4)$. Three retain the full simplex symmetry; the other two are exchanged by Frobenius and have stabilizer A_5 . The cycle argument needs only this unordered exotic pair. At two its graph slope has minimal polynomial $t^2 + t + 1$; at three every A_5 -stable graph is scalar. Both local forms admit primitive mixed-adjugate formulas, which produce $\Theta^4/4!$ as an integral product of divisor classes.

The quantum half uses a different invariant. We count primitive sixth-root eigenvalues of framed formal monodromy on the numerical small even quantum connection. Cai’s cubic calculation gives a nonzero count, and the projective-bundle formula doubles it on $X \times \mathbf{P}^1$. Iritani’s blowup comparison is additive for this count. Its center Novikov map can be noninjective; formal divisor tags restore injectivity and the divisor equation returns to the original specialization. Surface classification then shows that every center of dimension at most two contributes zero.

Local cofactor saturation forces the primitive minimal class into the integral divisor-product lattice; semisimple graph slopes give a sufficient mechanism. The second argument is a direct weak-factorization calculation with the framed quantum invariant. The special A_5 -geometry enters only in the first argument; the second applies to every smooth cubic threefold.

Relation with earlier rationality results

Clemens and Griffiths proved that every smooth complex cubic threefold is irrational by means of its intermediate Jacobian [4]. That obstruction does not by itself control a product with projective space, and stable rationality of smooth cubic threefolds remains open. Cai recently recovered irrationality from the primitive-sixth-root formal monodromy of the cubic quantum connection [3]. Iritani and Iritani–Koto prove the quantum D -module decompositions for blowups and projective bundles [8, 9]; Katzarkov–Kontsevich–Pantev–Yu identify the nef-canonical normal form used for low-dimensional centers [10]. Theorem 1.2 combines these inputs at the next dimension.

On the cycle side, Voisin reduces universal CH_0 -triviality of a cubic threefold to algebraicity of the primitive minimal class of its intermediate Jacobian [13]. Beckmann–de Gaay Fortman place such minimal classes in a general integral-Fourier framework [2]. For the very general cubic threefold the minimal class is not algebraic: every curve class on the intermediate Jacobian is an even multiple of it, so a very general cubic threefold admits no decomposition of the diagonal [5, Theorem 1.3 and Corollary 1.4]. What Voisin’s criterion leaves is the countable union of moduli subvarieties on which the class is algebraic; Theorem 3.8 places an explicit exceptional family inside that union. Here the six elliptic quotients supplied by Roulleau and the special period locus isolated by Hartlieb make the class explicit: the new point is its membership in the ordinary integral divisor-product lattice for every smooth member of the nonstandard A_5 -pencil.

Proof map

Section 2 constructs the rank-ten symplectic envelope and identifies the exotic cubic gluing. Section 3 proves the local divisor-product theorem and applies it to the primitive minimal class. Section 4 proves Theorem 1.2 from formal monodromy and weak factorization. The final section combines the two detectors and records the exact scope boundary.

2 The integral envelope and the exotic cubic sheet

Let Ω be the six conjugate D_5 -subgroups of A_5 , equivalently the six Sylow-5 subgroups with their normalizers. Put

$$A(\Omega) = \ker(\mathbf{Z}^\Omega \xrightarrow{\sum} \mathbf{Z}), \quad A(\Omega)^\vee = \operatorname{Hom}(A(\Omega), \mathbf{Z}).$$

The perfect evaluation pairing defines a unimodular alternating form on

$$\mathcal{H}_{\text{ref}}(\Omega) = A(\Omega)^\vee e \oplus A(\Omega)f, \quad \langle xe, yf \rangle = y(x).$$

This is the hyperbolic symplectic envelope of the oriented six-set.

Proposition 2.1 (Universal envelope). *The construction $\Omega \mapsto \mathcal{H}_{\text{ref}}(\Omega)$ is functorial for permutations of Ω . Rationally it is the unique rank-ten symplectic A_5 -module whose two Lagrangians are copies of the standard five-dimensional constituent and whose pairing restricts to the root-weight evaluation pairing.*

Proof. The permutation lattice and its augmentation kernel are functorial. The evaluation pairing between $A(\Omega)$ and its dual is perfect, so the displayed alternating form is unimodular. Over $\mathbf{Q}$, the augmentation module is absolutely irreducible and its invariant symmetric form is unique up to scale. Schur’s lemma therefore writes every symplectic module with two such constituents as $A(\Omega)_{\mathbf{Q}} \otimes U$, where U is a symplectic plane. Requiring the two integral Lagrangians to pair perfectly fixes the root-weight lattice above. $\square$

The finite golden orientation enters at the prime two. Let

$$H_2 = \operatorname{Aug}(\mathbf{F}_2^\Omega) / \langle \mathbf{1} \rangle.$$

Then H_2 is simple and $\operatorname{End}_{A_5}(H_2) = \mathbf{F}_4$. A normalized conference operator acts integrally

on $\mathbf{Z}^\Omega$. After labeling Ω , one switching representative is

$$B = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & -1 & -1 \\ 1 & 1 & 0 & -1 & 1 & -1 \\ 1 & 1 & -1 & 0 & -1 & 1 \\ 1 & -1 & 1 & -1 & 0 & 1 \\ 1 & -1 & -1 & 1 & 1 & 0 \end{pmatrix}, \quad B^2 = 5I.$$

Passing to the canonical saturation

$$D_6^\vee = \mathbf{Z}^\Omega + \mathbf{Z} \frac{\mathbf{1}}{2}, \quad \varphi = \frac{I + B}{2},$$

gives $\varphi^2 - \varphi - 1 = 0$. Its reduction on the commutator heart is one of the two primitive scalars $\omega, \omega^2 \in \mathbf{F}_4$, and replacing B by $-B$ applies Frobenius.

Proposition 2.2 (Golden selection of the exotic pair). *The two orientations of the conference class determine the two primitive scalars in $\text{End}_{A_5}(H_2) = \mathbf{F}_4$. Under orientation reversal,*

$$B \mapsto -B, \quad \omega \mapsto \omega^2.$$

Thus the golden-orientation torsor is canonically the Frobenius torsor of the exotic two-primary principal gluings.

Proof. Every column of $(I + B)/2$ has the same half-integral class modulo $\mathbf{Z}^\Omega$, so $D_6^\vee$ is the least stable over-lattice. The commutator submodule of $D_6^\vee/2D_6^\vee$ is the four-dimensional six-point heart and its quotient is a trivial $\mathbf{F}_4$ -line. The extension has no invariant $\mathbf{F}_4$ -line, so the restriction of $\bar{\varphi}$ to the heart is primitive. Finally $(I - B)/2 = 1 - \varphi$, and in characteristic two $1 - \omega = \omega^2$. $\square$

We now recall the geometric source of the same lattice. Let S be the Fano surface of a smooth member of the A_5 -pencil and $(J, \Theta) = \text{Alb}(S)$ its intermediate Jacobian. Each of the six D_5 -subgroups gives an elliptic quotient $q_H : J \rightarrow E_H$ and its Rosati dual $i_H : E_H \rightarrow J$.

Proposition 2.3 (Six-axis polarization). *After coherent identification of the elliptic factors, the Rosati Gram matrix of the six maps is*

$$G_6 = 6I_6 - J_6.$$

Any five axes therefore give an isogeny $E^5 \rightarrow J$ whose pulled-back polarization has Smith type $(1, 6, 6, 6, 6)$.

Proof. We recall the numerical passage from Rouleau's genus-two configuration to the six elliptic fibres. Let $D_g \subset S$ be the curve indexed by an involution $g \in A_5$, put $D_{\text{tot}} = \sum_g D_g$, and, for a dihedral subgroup H of order ten, put

$$F_H = \sum_{g \in H, g^2=1} D_g.$$

Rouleau proves that F_H is a fibre of $S \rightarrow E_H$ [12, Theorem 11(D)], so $F_H^2 = 0$. His intersection table has $D_g^2 = -4$, and distinct curves meet with multiplicity two or one when the product of their labels has order three or five, respectively. Among the 105 unordered pairs of involutions, the product orders two, three, and five occur 15, 30, 60 times. Hence

$$D_{\text{tot}}^2 = 15(-4) + 2(30 \cdot 2 + 60 \cdot 1) = 180.$$

Every involution belongs to two of the six dihedral subgroups, and therefore $\sum_H F_H = 2D_{\text{tot}}$. Two-transitivity on the six subgroups now gives

$$30(F_H \cdot F_{H'}) = (2D_{\text{tot}})^2 = 720, \quad F_H \cdot F_{H'} = 24 \quad (H \neq H').$$

After coherent identification of the generic non-CM elliptic quotients, two-transitivity makes the Rosati matrix have diagonal d and off-diagonal m . Put

$$D_H = q_H^*[0] \in \text{NS}(J), \quad N_H = i_H q_H \in \text{End}(J).$$

Then $D_H|_S = F_H$, and N_H is the Rosati endomorphism attached to D_H . Coherent identification gives $q_H i_H = [d]$ and $q_H i_{H'} = [m]$ for $H \neq H'$. The invariant sum of the six axis maps vanishes because the five-dimensional constituent has no trivial summand; composing with q_H gives $d + 5m = 0$. The minimal-class identity $[S] = \Theta^3/3!$, together with the polarized Riemann–Roch trace formula

$$\frac{\alpha\beta\Theta^{g-2}}{(g-2)!} = \text{tr}(N_\alpha) \text{tr}(N_\beta) - \text{tr}(N_\alpha N_\beta),$$

obtained by polarizing the reduced-norm Riemann–Roch polynomial [6, Theorem 4.1], gives

$$\text{tr}(N_H) = d, \quad \text{tr}(N_H N_{H'}) = m^2, \quad F_H F_{H'} = d^2 - m^2.$$

Thus $d^2 - m^2 = 24$. The positive solution is $(d, m) = (5, -1)$, hence $G_6 = 6I - J$. A five-by-five minor has Smith form $(1, 6, 6, 6, 6)$. $\square$

Proposition 2.4 (Principal gluing packet). *The A_5 -stable two-primary principal kernels of the six-axis source are parametrized by*

$$\mathbf{P}^1(\mathbf{F}_4) = \mathbf{P}^1(\mathbf{F}_2) \sqcup \{\omega, \omega^2\}.$$

The three $\mathbf{F}_2$ -rational kernels are S_6 -stable. Each of the other two has stabilizer exactly A_5 , and the outer normalizer exchanges them. The intermediate Jacobian of the nonstandard A_5 -cubic component uses the latter pair. Every A_5 -stable three-primary kernel is a scalar graph; the $\Gamma_0(3)$ multiplicity monodromy selects one of them.

Proof. At two the coefficient discriminant is the simple four-dimensional $\mathbf{F}_2 A_5$ -module H_2 , whose commutant is $\mathbf{F}_4$. The full discriminant is $H_2 \otimes_{\mathbf{F}_2} \mathbf{F}_2^2$. An A_5 -stable simple half is therefore a one-dimensional $\mathbf{F}_4$ -subspace of $\mathbf{F}_4^2$, and every such half is maximal isotropic for the tensor product of the coefficient form with the alternating multiplicity form. This gives $\mathbf{P}^1(\mathbf{F}_4)$.

Here is the polarization check explicitly. Identify the coefficient heart with $\mathbf{F}_4^2$ and put

$$b(x, y) = \text{Tr}_{\mathbf{F}_4/\mathbf{F}_2} \det(x, y).$$

On two multiplicity copies use

$$\Omega((x, y), (x', y')) = b(x, y') + b(y, x').$$

Every scalar $a \in \mathbf{F}_4$ is self-adjoint for b , since $b(ax, y) = b(x, ay)$. Thus the vertical line and all four graphs Γ_a are isotropic of half dimension. The five module-theoretic possibilities are exactly the five polarized gluings; no further quadratic condition removes the exotic pair.

The simplex group acts on both multiplicity copies of the coefficient heart over $\mathbf{F}_2$. Hence the three lines defined over $\mathbf{F}_2$ retain S_6 . The stabilizer of an exotic graph is equally intrinsic. A permutation that fixes Γ_ω commutes with ω , hence is $\mathbf{F}_4$ -linear on the two-dimensional $\mathbf{F}_4$ -space H_2 . If its determinant is $d \in \mathbf{F}_4^\times$, preservation of b says

$$\mathrm{Tr}_{\mathbf{F}_4/\mathbf{F}_2}(dz) = \mathrm{Tr}_{\mathbf{F}_4/\mathbf{F}_2}(z) \quad (z \in \mathbf{F}_4).$$

The trace pairing is nondegenerate, so $d = 1$. It therefore lies in $\mathrm{SL}_2(4)$, of order 60. The given A_5 already has that order and lies in the stabilizer, proving equality. The odd coset in its order-120 normalizer is semilinear and applies field Frobenius, so it exchanges ω and ω^2 . Were the generic cubic kernel $\mathbf{F}_2$ -rational, its principal polarization would inherit the faithful S_6 -action. Strong Torelli identifies $\mathrm{Aut}(J, \Theta)$ with $\mathrm{Aut}(X) \times \langle -1 \rangle$. Projection of this S_6 to $\mathrm{Aut}(X)$ remains faithful: its kernel would be a normal subgroup of S_6 of order at most two, and no such nontrivial subgroup exists. This contradicts the generic automorphism group A_5 on the nonstandard component [7, Section 5]. Thus the generic kernel is exotic. The three rational kernels and the two exotic kernels are distinguished by their stabilizers under the fibrewise A_5 -action; they are therefore distinct monodromy-stable subsets of the five-element gluing packet. The geometric kernel is a finite sub-local-system, so generic exoticity and the connectedness of the smooth locus force it to remain in the exotic pair on every fibre.

At three the coefficient heart H_3 is simple and $\mathrm{End}_{A_5}(H_3) = \mathbf{F}_3$. Hence its A_5 -stable halves are the four graphs parametrized by $\mathbf{P}^1(\mathbf{F}_3)$, all scalar in a suitable graph chart. $\square$

Proposition 2.2 labels the two exotic members by the finite golden orientation.

For the cycle-theoretic argument, the unordered pair is enough. Both members act on the heart with the same squarefree minimal polynomial $t^2 + t + 1$. The golden orientation supplies the finer marked identification with one member of the pair, but Theorem 3.8 uses only the conclusion that the geometric kernel is not one of the three $\mathbf{F}_2$ -rational gluings. At three it uses only scalarity of the possible A_5 -stable graphs, not the finer monodromy selection.

Hartlieb proves that its intermediate-Jacobian image is a one-dimensional special subvariety [7, Proposition 5.7]. Since the cubic period map is a locally closed embedding, the resulting smooth family is non-isotrivial. Only this landing and nonconstancy are used below; no identification of compactified parameter curves is needed for the separation theorem.

3 The primitive minimal class

Let (A, Θ) be a principally polarized abelian variety of dimension g . Write

$$\gamma_A = \frac{\Theta^{g-1}}{(g-1)!} \in H^{2g-2}(A, \mathbf{Z}).$$

The issue is not the rational algebraicity of γ_A , but its membership in the integral lattice generated by ordinary products of divisor classes. For an isogeny from an elliptic power, normalized Riemann–Roch turns this question into a cofactor problem. If P_p is the local lattice generated by mixed adjugates of descended coefficient divisors, then

$$\mathrm{adj}(G_p) \in P_p \text{ for every prime } p \implies \gamma_A \text{ is divisor-generated.}$$

The results below give two structural ways to force this local cofactor saturation: semisimple graph slopes and scalar Jordan blocks. The cubic family uses the former at two and the latter at three.

We use one standard graph chart explicitly. Let E be an elliptic curve, give E^g the coefficient polarization pI_g , and choose symplectic coordinates on $E[p]$. A symmetric matrix $T \in \text{Sym}_g(\mathbf{F}_p)$ defines the maximal isotropic subgroup

$$K_T = \{(x, Tx) : x \in \mathbf{F}_p^g\} \subset E[p]^g.$$

The quotient E^g/K_T carries the descended principal polarization.

Lemma 3.1 (The graph descent congruence). *Choose a symmetric integral lift $\tilde{T}$ of T . For $D \in \text{Sym}_g(\mathbf{Z}_p)$, the coefficient divisor pD on the source descends through the graph quotient whenever*

$$D\tilde{T} - \tilde{T}D \equiv 0 \pmod{p}.$$

In particular, on a summand on which the graph slope is scalar modulo p , every block-supported coefficient in $p\text{Sym}_d(\mathbf{Z}_p)$ descends.

Proof. Write the source homology as $L_e \oplus L_f$, with its standard alternating coordinates. The quotient homology is generated over the source by

$$p^{-1}(x, \tilde{T}x), \quad x \in L.$$

The alternating form of the coefficient divisor pD pairs an integral generator with each displayed graph generator integrally. On two graph generators its value is

$$\frac{1}{p}x^\dagger(D\tilde{T} - \tilde{T}D)y.$$

The stated congruence is therefore exactly the remaining integrality condition. If the slope is scalar on a block, the commutator vanishes there modulo p . $\square$

We first isolate the normalization that will be used throughout. If $C_1, \dots, C_d$ are symmetric endomorphisms of a free module, write

$$\text{Mdet}(C_1, \dots, C_d) = [t_1 \cdots t_d] \det\left(\sum t_i C_i\right)$$

and define $\text{Madj}(C_1, \dots, C_{d-1})$ in the same way from the adjugate. Thus these are coefficient operators, not derivatives divided by a factorial.

Lemma 3.2 (Primitive mixed coefficients). *On $\text{Sym}_d(\mathbf{Z})$, the mixed determinant has image $\mathbf{Z}$, and the entries of the mixed adjugates span $\text{Sym}_d(\mathbf{Z})$. More precisely,*

$$\text{Mdet}(E_{11}, \dots, E_{dd}) = 1,$$

while the coefficient of $\prod_{r \neq i} t_r$ in $\text{adj}(\sum_{r \neq i} t_r E_{rr})$ is E_{ii} , and inserting $E_{ij} + E_{ji}$ together with the E_{rr} , $r \neq i, j$, gives $-E_{ij} - E_{ji}$.

Proof. The first two assertions are the diagonal monomials in the Leibniz formulas for the determinant and its cofactors. For the last assertion the only surviving permutations use the transposition (ij) , whose sign is minus. Every displayed coefficient is therefore ± 1 . In particular neither a factorial nor the usual factor two for an off-diagonal quadratic term occurs. $\square$

The determinant is also the polarization dictionary. If a polarization on an elliptic power has coefficient matrix G , normalized Riemann–Roch identifies its top intersection polynomial with $\det G$ [6, Theorem 4.1 and Corollary 4.2]. Polarizing that identity shows that a product of $g - 1$ coefficient divisors is represented by the corresponding mixed adjugate. In particular, the pullback of the primitive minimal class is represented by $\text{adj}(G)$. This is why the coefficient-one statements in Lemma 3.2 are the exact integral normalization needed below.

Theorem 3.3 (Local cofactor saturation). *Let the coefficient form of an elliptic-power isogeny have, over $\mathbf{Z}_p$, an orthogonal Jordan decomposition*

$$G \simeq \bigoplus_a p^a B_a,$$

where each B_a is symmetric and unimodular of rank d_a . Suppose the pullback of the divisor lattice on the quotient contains every block-supported coefficient in

$$\bigoplus_a p^a \operatorname{Sym}_{d_a}(\mathbf{Z}_p).$$

Then $\operatorname{adj}(G)$ belongs to the image of the local product of $g - 1$ descended divisors.

Proof. For a target block a , fill every other block b with its primitive degree- d_b mixed determinant and fill block a with the primitive degree- $(d_a - 1)$ mixed adjugates from Lemma 3.2. The scalar coefficient is

$$\left(\prod_{b \neq a} p^{bd_b} \det B_b \right) p^{a(d_a - 1)} \operatorname{adj}(B_a),$$

which is exactly the a -block of $\operatorname{adj}(G)$. Disjoint block support introduces no multinomial coefficient. Summing over the target blocks gives the full cofactor. $\square$

Theorem 3.4 (Semisimple graph slopes). *Let p be a prime and let $T \in \operatorname{Sym}_g(\mathbf{F}_p)$. Form the principally polarized graph quotient above. If the minimal polynomial of T is squarefree, then*

$$\gamma_A \in \operatorname{im}(\operatorname{Sym}^{g-1} \operatorname{NS}(A) \longrightarrow H^{2g-2}(A, \mathbf{Z})).$$

Proof. Pass to a finite unramified extension $O/\mathbf{Z}_p$ over which the squarefree minimal polynomial splits. The self-adjoint eigenspaces are orthogonal, nondegenerate, and lift integrally without division by two. On each lifted scalar block, Lemma 3.1 supplies the full coefficient lattice $p \operatorname{Sym}_{d_\lambda}(O)$, so Theorem 3.3 supplies the cofactor. Formation of the divisor-product image commutes with $O/\mathbf{Z}_p$, and faithful flatness descends membership to $\mathbf{Z}_p$. $\square$

Corollary 3.5 (Unimodular coefficient form). *Suppose that, over $\mathbf{Z}_p$, the coefficient lattice is an orthogonal sum $U_0 \perp pB$, where U_0 and B are unimodular. Assume that the principal kernel is trivial on U_0 and is a graph on the defect block whose slope is self-adjoint for B and has squarefree minimal polynomial modulo p . Then the pullback of the primitive minimal class belongs to the local lattice generated by products of $g - 1$ descended divisors.*

Proof. The unit block descends. Over an unramified splitting ring, the eigenspaces in the defect block are B -orthogonal, nondegenerate, and integral. Lemma 3.1 gives the block-supported lattice $p \operatorname{Sym}(O)$. Apply Theorem 3.3 and descend by faithful flatness as above. $\square$

Corollary 3.6 (Squarefree gluing criterion). *Let $f : E^g \rightarrow (A, \Theta)$ be a coefficient isogeny from an elliptic power to a principally polarized abelian variety. Suppose that, for every prime $p \mid \deg f$, its local coefficient form and principal kernel satisfy the hypotheses of Corollary 3.5. Then*

$$\gamma_A \in \operatorname{im}(\operatorname{Sym}^{g-1} \operatorname{NS}(A) \longrightarrow H^{2g-2}(A, \mathbf{Z})).$$

Proof. Corollary 3.5 gives membership after tensoring with $\mathbf{Z}_p$ at every prime dividing $\deg f$. At every other prime, f identifies the integral divisor and cohomology lattices. The divisor-product cokernel is finitely generated, so these local memberships are equivalent to integral membership. $\square$

The scalar version of the same mixed-adjugate construction applies to the three-primary block of the six-axis polarization. We can therefore use the two local forms isolated in Section 2.

Lemma 3.7 (Local chart for the six-axis quotient). *Let $G = 6I_5 - J_5$. At each prime $p \mid 6$, the coefficient lattice has an orthogonal Jordan decomposition*

$$(\mathbf{Z}_p^5, G) \simeq U_0 \perp pU_1,$$

where U_0 has rank one and is unimodular and U_1 has rank four and is unimodular. The principal quotient is trivial on U_0 . On U_1/pU_1 , its graph slope has minimal polynomial $t^2 + t + 1$ at $p = 2$, while at $p = 3$ every A_5 -stable graph slope is scalar on the simple coefficient constituent.

Proof. The Smith form of G is $(1, 6, 6, 6, 6)$, but here the orthogonal chart can also be written down. Let $e_1, \dots, e_5$ be the standard basis. Since $\langle e_1, e_1 \rangle = 5$ and $\langle e_1, e_j \rangle = -1$, the vectors

$$v_j = e_j + \frac{1}{5}e_1 \quad (2 \leq j \leq 5)$$

span the orthogonal complement of e_1 over $\mathbf{Z}_p$. In this basis its Gram matrix is

$$\frac{6}{5}(5I_4 - J_4).$$

For $p = 2, 3$, both 5 and $\det(5I_4 - J_4) = 5^3$ are units. We have therefore exhibited a unimodular line orthogonal to p times a unimodular rank-four block. The unit block has no discriminant and hence no gluing choice.

At two the discriminant heart is the six-point module H_2 , and Proposition 2.2 identifies the selected endomorphism with $\omega \in \mathbf{F}_4$; its polynomial is $t^2 + t + 1$. At three the coefficient heart is simple with scalar commutant, so every A_5 -stable half is a scalar graph. The monodromy-invariant line identifies which scalar graph occurs in the family but is not needed for primitivity. $\square$

Theorem 3.8 (Minimal class for the exotic cubic family). *For every smooth member X_b of the nonstandard A_5 -cubic pencil, with intermediate Jacobian (J_b, Θ_b) ,*

$$\frac{\Theta_b^4}{4!} \in \text{im}(\text{Sym}^4 \text{NS}(J_b) \longrightarrow H^8(J_b, \mathbf{Z})).$$

Proof. The five-axis source has polarization type $(1, 6, 6, 6, 6)$, so only the primes two and three can contribute an integral defect. At two the exotic graph slope acts on the simple heart through $\omega \in \mathbf{F}_4$; its minimal polynomial $t^2 + t + 1$ is squarefree. Theorem 3.4 via Corollary 3.5 therefore gives the primitive cofactor on the rank-four defect block. The rank-one unit block has trivial kernel, so its primitive coefficient divisor descends. Filling the unit block by its determinant when the defect block is the adjugate target, and filling the defect block by its determinant when the unit block is the target, constructs the cofactor of the full five-dimensional form. At three every A_5 -stable gluing is scalar on the simple coefficient block, and the same blockwise construction applies. Away from two and three the source and quotient lattices are unimodular. Local membership at every prime is equivalent to integral membership in the divisor-product lattice.

Lemma 3.7 identifies the actual geometric lattice, not merely an abstract gluing with the same Smith form. The graph kernel is a finite sub-local-system on each connected smooth component of the pencil, so the same local presentation holds on every fibre. At a fibre with additional Néron–Severi classes the divisor-product image can only increase. $\square$

Voisin identifies this minimal class as the exact obstruction relevant to cubic threefolds.

Corollary 3.9 (Universal CH_0 -triviality). *Every smooth member of the nonstandard A_5 -cubic pencil is universally CH_0 -trivial.*

Proof. By Theorem 3.8, the primitive minimal class of the intermediate Jacobian is algebraic. Voisin's Corollary 4.4 [13] says that, for a smooth complex cubic threefold, this is equivalent to universal CH_0 -triviality. $\square$

The argument is fibrewise. It constructs neither a horizontal minimal cycle over the pencil nor a relative decomposition of the diagonal. Neither is needed for Corollary 3.9.

4 Framed formal monodromy

The obstruction used here belongs to the small even quantum connection itself. We follow it through each blowup in a weak factorization, without passing to equivalence classes of quantum atoms. This avoids a subtlety: a formal bulk gauge is coefficientwise Laurent in the loop coordinate, but need not specialize to a meromorphic gauge at a nearby analytic point.

Let T be a smooth projective variety. Write

$$\Lambda_T = \mathbf{C}[[\mathrm{NE}_{\mathrm{num}}^{\mathbf{Z}}(T)]]$$

for the completed monoid ring of the image of effective homology classes in the numerical curve lattice

$$N_1(T) = \mathrm{im}(H_2(T, \mathbf{Z}) \longrightarrow \mathrm{Hom}(\mathrm{NS}(T)/\mathrm{tors}, \mathbf{Z})).$$

Thus $N_1(T)$ is torsion-free and is separated by integral divisors. Let $\mathcal{K}_T$ be an algebraic closure of its fraction field. The loop coordinate z remains independent: in particular, $\mathcal{K}_T$ contains no chosen root of z . Apply the Levelt–Turrittin construction to the small even quantum connection over $\mathcal{K}_T((z))$. Its formal monodromy is the action of one turn on the original z -disc. Thus it includes both regular-singular monodromy and the deck permutation of ramified exponential factors.

Definition 4.1. Let $M_f(T)$ be this framed formal-monodromy operator, and put

$$\nu_6(T) = \mathrm{mult}_{e^{\pi i/3}} M_f(T) + \mathrm{mult}_{e^{-\pi i/3}} M_f(T).$$

The multiplicities are algebraic multiplicities in the characteristic polynomial.

The frame matters. Adjoining roots of Novikov monomials is harmless, but adjoining a root of z would replace the original loop and could erase the deck part of the operator.

We first record two formal facts used in the comparison theorems. They also fix the completion convention in the proof.

Lemma 4.2 (Coefficientwise base change). *Let D be a formal differential module over $k((z))$, and let B be a complete separated I -adic k -algebra on which d/dz is zero. Scalar extension in the coefficient direction preserves the characteristic polynomial of framed formal monodromy. The same holds after a connection gauge whose entries use only integral powers of z .*

Proof. Choose a Level–Turrittin solution algebra R for D with differential constants k . At every finite level, the constants of $R \otimes_k B/I^N$ are B/I^N : expansion in a k -basis reduces this to $R^{d/dz} = k$. A fundamental solution matrix then identifies the horizontal module with

$$\mathrm{Sol}(D) \otimes_k B/I^N.$$

Passing to the inverse limit gives $\mathrm{Sol}(D) \hat{\otimes}_k B$, with monodromy $M_f(D) \otimes 1$. An integral- z gauge is single-valued on the original disc and conjugates this action. $\square$

Lemma 4.3 (Formal base shift). *Suppose a pullback of the even quantum connection of T is defined over a complete filtered coefficient ring B , and its bulk coordinate has the form*

$$a_0 \mathbf{1} + a_2^\circ + \eta, \quad a_2^\circ \in H^2(T) \otimes F^0 B, \quad \eta \in F^1 B \hat{\otimes} H^{\mathrm{ev}}(T).$$

Assume that a_0 and a_2° are constant in the positive filtration. Its framed formal-monodromy characteristic polynomial equals that of the small connection after the divisor substitution

$$Q^d \longmapsto e^{\langle a_2^\circ, d \rangle} Q^d. \quad (4.1)$$

Proof. The string equation removes $a_0 \mathbf{1}$ by a scalar exponential in z^{-1} ; this is single-valued and has trivial framed monodromy. The divisor equation gives (4.1). It remains to remove η , including its component in $H^2(T)$.

Modulo $F^N B$, the flat bulk equations

$$\partial_{\eta_i} G = -z^{-1}(\phi_i \star_\eta) G, \quad G|_{\eta=0} = 1,$$

have an invertible solution with bounded Laurent order in z . These solutions are compatible in N . Their inverse limit may have negative z -orders unbounded with the filtration degree, but every coefficient uses only integral powers of z . Lemma 4.2, applied at each finite level and then in the inverse limit, proves the assertion. No evaluation of this gauge at a nonnilpotent analytic bulk point is used. $\square$

We now apply Iritani’s quantum blowup theorem and the Iritani–Koto projective-bundle theorem. Both are stated over completed homological Novikov rings.

Lemma 4.4 (Numerical Novikov base change). *The quotient from the effective homology monoid to its image in $N_1(T)$ extends continuously to completed monoid rings, and the quantum connection and the two comparison theorems base-change along it.*

Proof. Filter both monoids by intersection with an ample divisor. Effective homology classes below a fixed cutoff form a finite set, as follows from the finite-type Chow varieties of cycles of bounded degree. The quotient is therefore defined on every truncated monoid ring and passes to the inverse limit. The quantum products descend by finite sums of coefficients with the same numerical class. Pushforward of curves, pairing with $c_1(N_{C/T})$, and the derivations $Q \partial_Q$ depend only on numerical classes. Since the comparison theorems provide both a map and its inverse, they remain inverse after this base change. $\square$

Proposition 4.5 (Framed operation formulas). *Let $V \rightarrow T$ be a vector bundle of rank $r \geq 2$. Then*

$$\nu_6(\mathbf{P}_T(V)) = r \nu_6(T). \quad (4.2)$$

Let $\tilde{T} = \mathrm{Bl}_C T$, where $C \subset T$ is smooth of codimension $c \geq 2$. In Iritani’s comparison field one has

$$\nu_6(\tilde{T}) = \nu_6(T) + \sum_{j=0}^{c-2} \nu_6(C; \chi_j), \quad (4.3)$$

where χ_j is the numerical Novikov specialization of the j -th center summand. The endpoint terms in (4.3) are the intrinsic small invariants; the center terms are computed after the displayed specializations.

Proof. Iritani's Theorem 5.18 gives a formal coordinate isomorphism and a quantum- D -module isomorphism

$$\mathrm{QDM}(\tilde{T}) \cong \mathrm{QDM}(T) \oplus \mathrm{QDM}(C)^{\oplus(c-1)}$$

over a Laurent extension in the exceptional Novikov variable [8, Theorem 5.18]. The isomorphism and its inverse use only integral powers of z , and the mirror coordinates are independent of z . The center base shift is controlled before passage to the final Laurent chart. Put $u = q^{-1/(c-1)}$ and use the reconstruction coordinates of [8, Section 5.8.2]:

$$\varsigma_j = \varsigma_j^\circ + s_j, \quad \varsigma_j^\circ = -(c-1)\lambda_j + h_{C,j} + O(u). \quad (4.4)$$

For the j -th target summand, let R_j be the image of $Q_C^d \mapsto Q^{i_*d} u^{\rho_C \cdot d}$, together with u and s_j , and set

$$J_j = (u, \{Q^{i_*d} u^{\rho_C \cdot d} : d \neq 0\}, \text{components of } s_j) \subset R_j, \quad B_j = \varprojlim_N R_j / J_j^N.$$

The center connection is defined over this image ring by [8, Remark 5.6]. Formula (4.4), together with (5.45), (5.47), and the initial asymptotics (5.27)–(5.30), shows that after the unit term $-(c-1)\lambda_j$ and fixed divisor $h_{C,j}$ are removed, its target bulk coordinate lies in $J_j H^*(C)$. The noninjective center monomial map merely replaces the source ring by its image.

First normalize the target connection over $\mathbf{C}((u))$ by the single-valued string scalar for $-(c-1)\lambda_j$ and by the fixed divisor shift. The normalized connection descends to B_j . Apply Lemma 4.3 modulo every J_j^N , take the compatible inverse limit, and only then extend scalars to Iritani's Laurent comparison field and apply Ψ . The fixed-divisor substitution is a z -constant coefficient automorphism, so it does not change ν_6 ; it is well defined on the image because $h_{C,j}$ is a scalar multiple of ρ_C . Thus each center target summand has the framed monodromy of its specialized small connection. The ambient target uses $\tau = \tau^\circ + t$, with $\tau^\circ = O(q^{-1})$, and the same finite-quotient argument. Direct-sum additivity gives (4.3).

The endpoint map is faithful: on numerical curve lattices

$$\bar{\beta} \mapsto (\phi_* \bar{\beta}, -E \cdot \bar{\beta})$$

is injective because $N^1(\tilde{T}) = \phi^* N^1(T) \oplus \mathbf{Z}[E]$. Hence the Laurent extension preserves the intrinsic endpoint characteristic polynomials.

The projective-bundle proof is parallel, again in reconstruction coordinates. Put $u = q^{-1/r}$. For its j -th target summand let R_j be the image of $Q_T^d \mapsto Q^d u^{c_1(V) \cdot d}$, together with u and s_j , and set

$$J_j = (u, \{Q^d u^{c_1(V) \cdot d} : d \neq 0\}, \text{components of } s_j) \subset R_j, \quad B_j = \varprojlim_N R_j / J_j^N.$$

Proposition 5.6 and Section 5.8 of Iritani–Koto give

$$\varsigma_j = \varsigma_j^\circ + s_j, \quad \varsigma_j^\circ = r\lambda_j - \frac{2\pi i j}{r} c_1(V) + O(u) [9, \text{Proposition 5.6 and Section 5.8}].$$

Normalize the Laurent unit and fixed divisor terms first. The normalized connection descends to B_j ; apply Lemma 4.3 in its finite quotients, and then extend scalars to the

Laurent comparison field. Again the divisor substitution is a z -constant coefficient automorphism; it factors through the displayed u -exponent. This gives r copies of the intrinsic base module and proves (4.2). The base Novikov map is the embedding in [9, (1.1) and Remark 5.2]. The theorem assumes $V^\vee$ globally generated. Tensoring V by a sufficiently negative line bundle achieves this without changing $\mathbf{P}_T(V)$ [9, Remark 1.2]; the resulting divisor shift preserves framed monodromy by Lemma 4.3. All other Novikov extensions are in the coefficient direction. After the odd bulk variables vanish, pullback, pushforward, Fourier transformation, and multiplication by the characteristic-class factors all preserve cohomological parity. Hence both decompositions restrict to the even quantum connections used here. Formula (4.2) is tautological when $r = 1$. $\square$

The center map in (4.3) is not necessarily injective [8, (5.15)]. Thus generic vanishing for C does not by itself imply that its specialized term vanishes. The next lemma supplies the missing implication.

Definition 4.6. A monomial map $\chi : \Lambda_T \rightarrow A$ is *strictly Novikov-admissible* if A is a complete separated valued domain with domain associated graded, every monomial has nonzero image, and

$$v(\chi(Q^d)) = \ell(d), \quad (4.5)$$

where ℓ is positive and proper on the effective numerical monoid.

The center maps in (4.3) are strictly admissible. Give the exceptional Laurent variable valuation zero and use an ample divisor H on T . Then

$$v_H(Q^{i_*d} q^{-\rho_C \cdot d/(c-1)}) = (H|_C) \cdot d.$$

The restriction $H|_C$ is ample, which gives positivity and properness; the associated graded Laurent monoid ring is a domain.

Lemma 4.7 (Divisor tagging). *Let $\chi : \Lambda_T \rightarrow A$ be strictly Novikov-admissible. If the intrinsic small even quantum connection of T has no primitive-sixth framed monodromy, neither does its specialization along χ .*

Proof. Choose integral divisors $D_1, \dots, D_\rho$ whose pairings separate $N_1(T)$, and introduce formal variables $\mathbf{t} = (t_1, \dots, t_\rho)$. Define

$$\chi_{\mathbf{t}}(Q^d) = \chi(Q^d) \exp\left(\sum_i t_i (D_i \cdot d)\right). \quad (4.6)$$

This map from Λ_T to $A[[\mathbf{t}]]$ is injective. To prove this, take a nonzero series. Properness in (4.5) makes its lowest-valuation support finite. Its initial image is a finite combination of distinct exponential characters. After a generic substitution $t_i = a_i s$, their exponents are distinct; their derivatives at $s = 0$ form a Vandermonde matrix. The characters are therefore linearly independent. The domain hypothesis on the associated graded prevents cancellation of the initial terms.

Let $k_\chi = \text{Frac } \bar{A}$. After embedding the two fraction fields in a common algebraic closure, injectivity makes the $\chi_{\mathbf{t}}$ -module a scalar extension of the intrinsic module. On the other hand, the divisor equation identifies (4.6) with the degree-two formal bulk pullback of the χ -module over $B = k_\chi[[\mathbf{t}]]$. Apply the finite-quotient argument in Lemma 4.3 with $\eta = \sum_i t_i D_i$: its integral- z bulk gauge removes the positive divisor component. Hence vanishing over the intrinsic field implies vanishing after χ . $\square$

Proposition 4.8 (Low-dimensional vanishing). *Let T be a point, a smooth projective curve, or a smooth projective surface. For every strictly Novikov-admissible specialization χ ,*

$$\nu_6(T; \chi) = 0. \quad (4.7)$$

Proof. We first prove the intrinsic generic statement. If K_T is nef, the proof of [10, Claim 6.15] applies coefficient by coefficient over the small numerical Novikov ring. Set

$$\text{Gr} = \frac{\text{Deg} - \dim T}{2}, \quad g = \text{Gr} + \frac{1}{2}E,$$

where E is the identity in cohomological degrees of parity opposite to $\dim T$ and is zero in the other degrees. The operator g has integral eigenvalues. Coefficientwise in Q^β , multiplication by $c_1(T)$ sends

$$H^i(T) \longrightarrow H^{i+2-2c_1(T)\cdot\beta}(T).$$

If K_T is nef, then $c_1(T) \cdot \beta \leq 0$, so the Gr-weight rises by $1 - c_1(T) \cdot \beta \geq 1$. Conjugation by z^g therefore turns $\nabla + \frac{1}{2}E dz/z$ into a regular-singular connection with nilpotent residue. Undoing the half-parity correction permits only the monodromy eigenvalues 1 and -1 , never a primitive sixth root.

This treats curves of positive genus and minimal surfaces with nef canonical class. A point is immediate. The remaining minimal varieties are $\mathbf{P}^1$, $\mathbf{P}^2$, and ruled surfaces; (4.2) reduces them to a point or a curve. Every nonminimal surface is obtained from a minimal surface by point blowups, and (4.3) adds only point summands. Hence the intrinsic generic invariant vanishes in every case. Lemma 4.7 gives (4.7). $\square$

Proposition 4.9 (The cubic packet). *For every smooth cubic threefold X ,*

$$\nu_6(X) \geq 2. \quad (4.8)$$

Proof. In the basis $1, P, P^2, P^3$, Cai explicitly block-diagonalizes the small even quantum connection. The rank-two zero-exponential block has indicial polynomial

$$\rho^2 + \rho + \frac{5}{36},$$

with roots $-1/6$ and $-5/6$ [3, Section 3 and Proposition 6]. His gauge uses only integral powers of z . The two corresponding framed-monodromy eigenvalues are $e^{-\pi i/3}$ and $e^{\pi i/3}$. The numerical curve lattice of a cubic threefold has rank one, so the numerical-Novikov convention does not change the calculation. $\square$

Proof of Theorem 1.2. The projective-bundle formula and Proposition 4.9 give

$$\nu_6(X \times \mathbf{P}^1) = 2\nu_6(X) > 0. \quad (4.9)$$

Suppose that $X \times \mathbf{P}^1$ were rational. Weak factorization of a birational map to $\mathbf{P}^4$ expresses it as a sequence of blowups and blowdowns with smooth centers [1]. A nontrivial center in a smooth fourfold has dimension at most two. For each step, (4.3) and Proposition 4.8 show that the two endpoint values of ν_6 are equal. This integer equality is reversible, so it telescopes through either orientation of the factorization; each step may use its own comparison field. We obtain

$$\nu_6(X \times \mathbf{P}^1) = \nu_6(\mathbf{P}^4) = 0,$$

where the last equality follows from (4.2) over a point. This contradicts (4.9). $\square$

The argument is sharp for this invariant. In a factorization of the fivefold $X \times \mathbf{P}^2$, a center may have dimension three and may itself carry the cubic packet. The next stabilization therefore requires information about the packet's Stokes or Rees placement, not only its multiplicity.

5 The separation theorem

Proof of Theorem 1.1. Corollary 3.9 gives universal CH_0 -triviality of X_b . The projective bundle formula for Chow groups gives the same property for $X_b \times \mathbf{P}^1$. Theorem 1.2 proves that this fourfold is irrational. Non-isotriviality follows from the nonconstant intermediate-Jacobian period map on the A_5 -component. $\square$

The theorem separates two tests on one variety. The algebraic class $\Theta^4/4!$ removes the decomposition-of-diagonal obstruction. The primitive-sixth framed-monodromy multiplicity survives one stabilization and obstructs a rational factorization. The first mechanism is special to the exotic A_5 -family; the second holds for every cubic threefold. More generally, the local cofactor-saturation theorem applies beyond this family; this geometry is where the two mechanisms hold simultaneously.

The result stops after one stabilization. For $X \times \mathbf{P}^m$ with $m \geq 2$, weak-factorization centers may have dimension at least three and can carry primitive-sixth quantum packets. The low-dimensional vanishing in Proposition 4.8 no longer proves noncancellation. Likewise, the fibrewise minimal-class argument gives no relative universal cycle over the pencil. These are separate problems and are not implicit in the theorem above.

Thus the same smooth fourfolds pass the universal CH_0 -triviality test and fail the framed-quantum rationality test. Vanishing of the decomposition-of-the-diagonal obstruction does not predict rationality even after one explicit stabilization.

References

- [1] D. Abramovich, K. Karu, K. Matsuki, and J. Włodarczyk, *Torification and factorization of birational maps*, J. Amer. Math. Soc. 15 (2002), 531–572.
- [2] T. Beckmann and O. de Gaay Fortman, *Integral Fourier transforms and the integral Hodge conjecture for one-cycles on abelian varieties*, Compos. Math. 159 (2023), 1188–1213.
- [3] J. Cai, *The cubic threefold is symplectically irrational*, arXiv:2608.01577 (2026).
- [4] C. H. Clemens and P. A. Griffiths, *The intermediate Jacobian of the cubic threefold*, Ann. of Math. (2) 95 (1972), 281–356.
- [5] P. Engel, O. de Gaay Fortman, and S. Schreieder, *Matroids and the integral Hodge conjecture for abelian varieties*, arXiv:2507.15704v3 (2026).
- [6] N. Grieve, *Reduced norms and the Riemann–Roch theorem for abelian varieties*, New York J. Math. 23 (2017), 1087–1110; arXiv:1603.06425.
- [7] M. Hartlieb, *Special subvarieties in the locus of intermediate Jacobians of cubic threefolds*, Math. Z. 310 (2025), article 52; arXiv:2304.03214.
- [8] H. Iritani, *Quantum cohomology of blowups*, arXiv:2307.13555v3 (2025).
- [9] H. Iritani and Y. Koto, *Quantum cohomology of projective bundles*, arXiv:2307.03696v4 (2026).
- [10] L. Katzarkov, M. Kontsevich, T. Pantev, and T. Y. Yu, *Birational invariants from Hodge structures and quantum multiplication*, arXiv:2508.05105v2 (2026).

- [11] M. van der Put and M. F. Singer, *Galois theory of linear differential equations*, Grundlehren Math. Wiss. 328, Springer, 2003.
- [12] X. Roulleau, *Genus 2 curve configurations on Fano surfaces*, Comment. Math. Univ. St. Pauli 59 (2010), 51–64; arXiv:1002.4467.
- [13] C. Voisin, *On the universal CH_0 group of cubic hypersurfaces*, J. Eur. Math. Soc. 19 (2017), 1619–1653; arXiv:1407.7261.